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Claim Interpretation - Preambles

Virtual Instructor Led Training (VILT) HOPS

Approved by OPLA on February 7, 2018

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Objective

You will review the following:

- Overview of claim interpretation
- Review and discuss examples of how preambles limit a claim

Claim Interpretation

Clarity of claim terms in granted patents is improved when the interpretation of claim terms is explained in Office actions

- Early explanation of the Examiner's interpretation will allow the Applicant to clarify the meaning of a term, amend the claim, and/or provide a more effective response to any rejections, thus, leading to more efficient prosecution
- The prosecution record will provide a map for the public to understand the boundaries of the patent protection
- The Patent Trial and Appeal Board (PTAB) and courts will be informed as to what the Examiner and the Applicant understood the claims to mean

From an Examiner's Point of View

If the claims are correctly **analyzed** in a first Office Action, the next Office action should result in:

- less rework (no credit) needed in future Office actions; and
- the second, subsequent Office action is more likely to be a Final Office action or an allowance

From an Examiner's Point of View (*cont.*)

Detailed and correct **claim interpretation** in an Office action will benefit the Examiner

Applicant will understand the issues and respond appropriately by:

- Persuasively arguing against the position of the Office Action
- Amending/narrowing the claims to place the application in condition for allowance
- Abandoning the application

From Applicant's Point of View

Applicant benefits from detailed and correct claim interpretation in an Office action because it:

- Reduces guesswork in responding to the Office action
- Provides public notice as to how the claim was interpreted during examination
- Clearly sets forth the protection afforded the Applicant
- Provides a clear file history to prevent or reduce unnecessary litigation, interferences, reissues, and reexaminations, *inter partes* reviews, supplemental examinations, and post-grant reviews

Claim Interpretation (*cont.*)

- Claim terms are given their plain meaning in the art being examined unless it is inconsistent with the specification
- Claims must be interpreted in light of the specification **without** reading limitations from the specification into the claims

See: MPEP 2111.01

Claim Interpretation (*cont.*)

- Broadest **Reasonable** Interpretation (BRI) is always used to interpret claims under examination
 - BRI does not mean broadest *possible* interpretation
 - The interpretation of a term must be consistent with the specification and how the term is commonly used in the art (*MPEP 2111*)

Claim Interpretation (*cont.*)

Also, every word in a claim **must** be considered in judging the patentability of that claim against the prior art:

- No limitation, functional or otherwise, can be ignored

See: MPEP 2143.03

Preambles

Parts of a Claim

A claim has three (3) main parts:

- A **preamble** or introduction
- A **transitional phrase**:
 - comprising (open); or
 - consisting of (closed); or
 - consisting essentially of (open, but excludes materials or steps that materially affect the basic and novel characteristics of the invention)
- A **body** reciting the elements of the invention

See: MPEP 608.01(m) & 2111.03

Preamble

A Preamble is an introductory phrase of a claim and *might*:

- 1) Summarize the invention;
- 2) Summarize its relation to the prior art;
- 3) Summarize its intended use or properties; and
- 4) Constitute a limitation of the claimed device or process.

When the Preamble Limits Claim Scope

The determination of whether a preamble limits a claim is made on a case-by-case basis in light of the facts in each case and should be based on the facts of each case in light of:

- 1.The overall form of the claim
- 2.The invention as disclosed in the specification
- 3.The invention as illuminated in prosecution history

See: *MPEP 2111.02*

When the Preamble Limits Claim Scope (*cont.*)

There is no litmus test in determining when a preamble limits the scope of a claim.

See: *MPEP 2111.02*

When a Preamble *Will Likely* Limit a Claim

More specifically,

- The preamble is necessary to give “life, meaning and vitality” to the claim, i.e., words that add context for claim construction may limit the claim
- The preamble provides antecedent basis for a limitation in the body of the claim
- The preamble recites, and thereby limits, the structure of the claimed invention
- The preamble recitation is relied upon during prosecution to distinguish the claimed invention from the prior art

See: *MPEP 2111.02 & 2111.02(II)*

A “Jepson” Claim Example

Claim.

In a hollow bat having a small-diameter handle portion and a large-diameter impact portion, **an improvement comprising** an internal structural insert defining an annular gap with an inside wall of the impact portion of the bat and the impact portion elastically deflectable to close a portion of the annular gap and operably engage the insert.

See: MPEP 2129

When a Preamble *Is Not Likely* to Limit a Claim

- The body of the claim following the preamble is a self-contained description of the structure and does not depend on the preamble for completeness
- A preamble that describes the purpose or intended use of the claimed invention generally does not limit the claims
- The preamble merely extols benefits or features of the claimed invention and there is no clear reliance on those benefits or features as patentably significant
(e.g., preamble recites, "[a] head for a lacrosse stick *which provides improved handling and playing characteristics.*")

See: *MPEP 2111.02 & 2111.02(II)*

Questions to Ask to Help Determine Whether a Preamble Limits a Claim

- Does the preamble language provide antecedent basis for terms in the body of the claim?
 - If YES, then the preamble language will likely be a limitation of the claim
- Does the language in the body of the claim set out the complete invention?
 - If YES, then the claim preamble will likely not be a limitation of the claim

NOTE: A new use for an old product does not render the old product patentable

Questions to Ask in Determining Whether a Preamble Limits a Claim (*cont.*)

- Does the language of the preamble merely provide a statement of purpose or intended use?
 - If YES, then the preamble limits the claim only to the extent that the prior art must be capable of performing the purpose or intended use
- Is the preamble essential to understanding limitations or terms in the body portion of the claim?
 - If YES, then the preamble may be a structural limitation of the claim

Example 1: A Protective Cover for a Barbeque Grill Claims

1. A protective cover ***for use with a barbeque grill***, comprising:
 - a body portion;
 - a lower flap and an upper flap, said lower flap being attached to said body portion such that a pocket having an opening is formed, said upper flap being attached to said body portion such that said upper flap covers said opening;
 - at least one fastener arranged and configured for connecting said upper flap to said lower flap; and
 - wherein said pocket is arranged and configured for storing said cover when said cover is not in use.

Example 1:

Claims (*cont.*)

2. A method for storing a cover ***for a barbeque grill***, the cover including a pocket formed by an upper flap and a lower flap, the upper and lower flaps including fasteners, comprising the steps of:
turning the pocket inside out;
placing the cover inside the now inside out pocket; and
fastening the upper flap to the lower flap with the fasteners.

Example 1:

Questions and Answers

Q1: In order to anticipate claim 1 (or 2), must a reference disclose a cover that would be capable of covering a grill?

Yes

Q2: In order to anticipate claim 1 (or 2), must a reference disclose actually using the cover for a grill?

No

See: *MPEP 2111.02*

Example 1:

The Prior Art: Poncho Raincoat



Example 1: Questions and Answers (*cont.*)

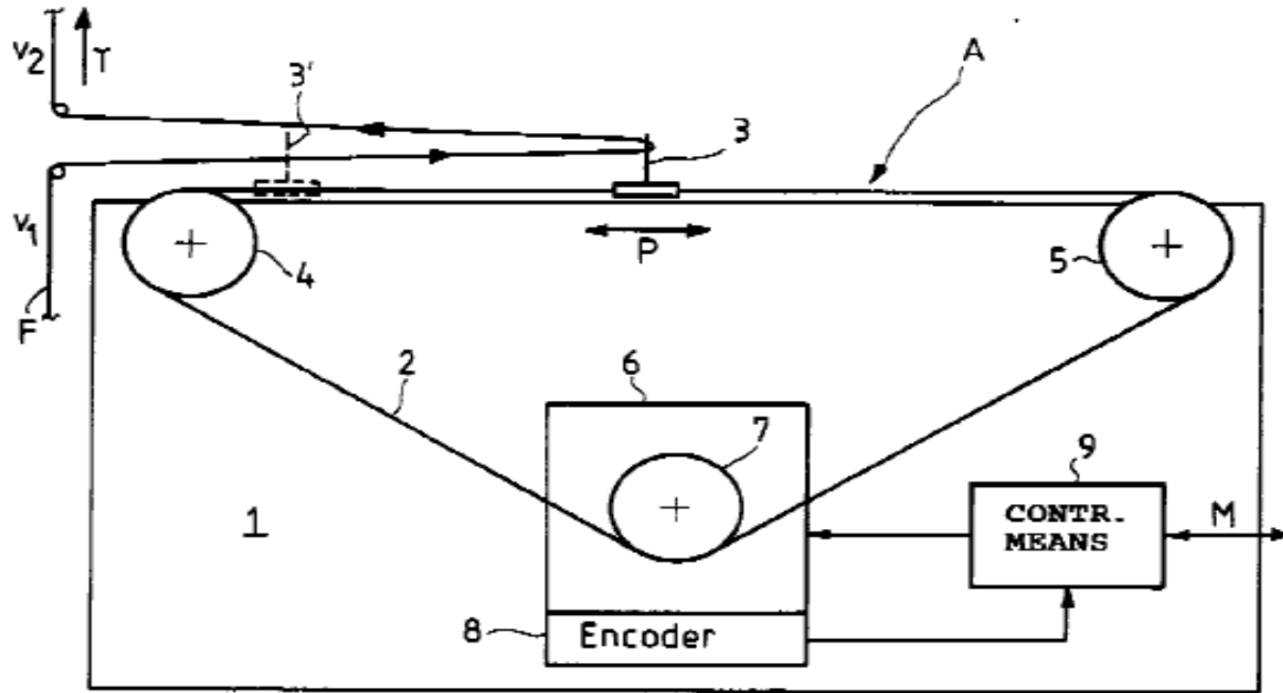
Q3: Could a rejection be made for claims 1 and/or 2 using the poncho raincoat as prior art?

Yes

Regarding claim 1, the poncho raincoat meets the structural requirements in claim 1 and is capable of being used as a protective cover for a barbeque grill.

Regarding claim 2, in achieving the stored configuration shown in the figure, one would appreciate that the claimed method steps are satisfied and is capable of being used as a cover for a barbeque grill.

Example 2: Device for Producing Effect Yarns



Example 2: Claim

An apparatus ***for producing effect yarns on process machines through a deliberate production of short lumps in the yarn through short-duration yarn overfeed***, comprising a device for lateral yarn deflection having a motor-driveable, servo-controlled yarn guide with an adjustable stroke, ***whereby a movement of the yarn guide in one direction results in a slight underfeed and movement in another direction results in a pronounced overfeed of the yarn.***

Example 2:

Questions and Answers

Q1: Does the claimed apparatus include a device in combination with the “process machines” recited in the preamble?

No

The preamble recitation is simply a statement of intended use.

Example 2:

Questions and Answers (*cont.*)

Q2: Does the functional recitation in the “whereby” clause impose any structural limitations on the claimed device?

Yes

The servo-controlled yarn guide is designed to perform the recited slight underfeed and pronounced overfeed; therefore, an anticipatory reference must be capable of being used to move a yarn guide in one direction to result in a slight underfeed and move in another direction to result in a pronounced overfeed.

Example 2:

Questions and Answers (*cont.*)

Q3: Must the reference provide any specific teaching that it could be used to result in the recited underfeed and overfeed?

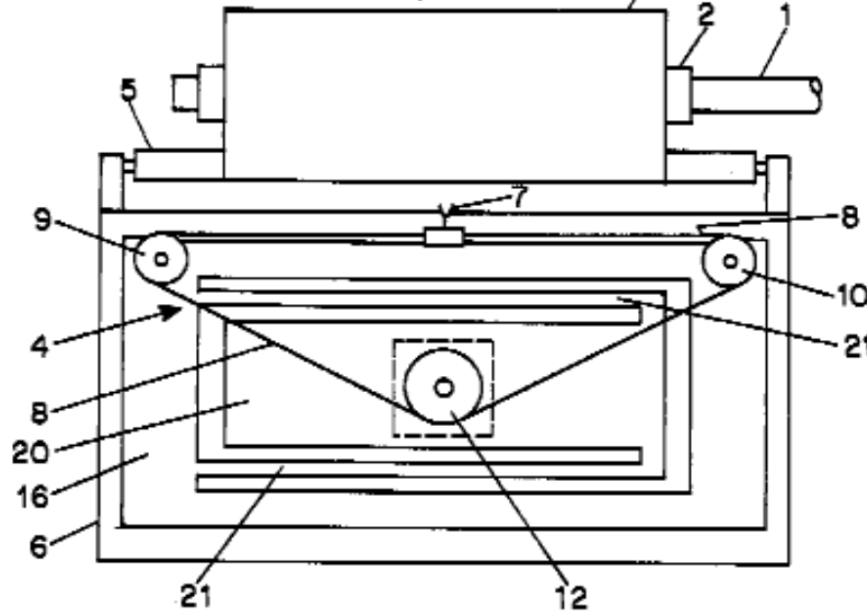
No

The Examiner need only determine that the reference structure could be so used, i.e., there must be nothing disclosed which would prevent the reference structure from being so used.

Example 2:

The Prior Art

Yarn guide 7 oscillates between pulleys 9 and 10 in response to drive wheel 12 operated by a servo motor:



Example 2:

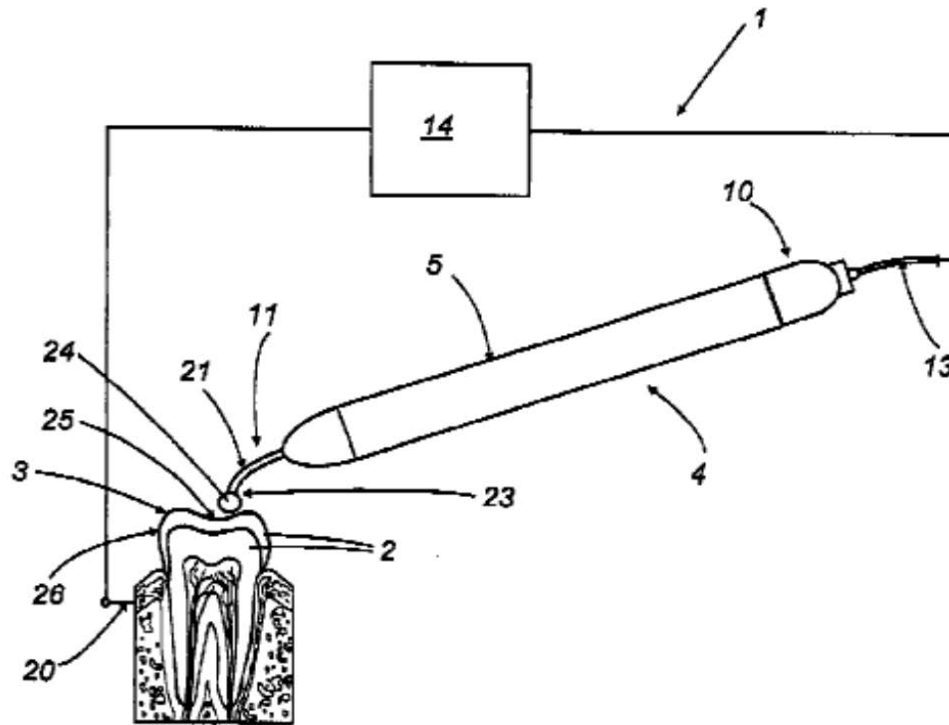
Questions and Answers (*cont.*)

Q4: Should a rejection be made using the prior art yarn guide?

Yes

It was noted that the preamble is merely directed to an intended use for the claimed yarn guide, thus failing to add any structure. The prior art yarn guide is capable of moving as claimed in the whereby clause. There is nothing in the prior art arrangement which would prevent such movement as stated in the whereby clause.

Example 3: Device for Preparing the Hard Structures of Teeth



Example 3:

Claim

A device ***for preparing the hard structures of teeth for the application of dental restorative materials***, comprising at least one handpiece connected to a control power circuit ***for applying an electrical field to at least one layer of adhesive substance applied at least to an affected zone of the hard structure of the tooth previously treated to remove the layer of tissue affected by the disorder, so as to apply the restorative material at least on the zone covered by the adhesive substance*** and an electrode connected to the control power circuit and ***designed to be placed at a point on the patient's body and operating in conjunction with the handpiece.***

Example 3:

Questions and Answers

Q1: Must a reference disclose applying an electrical field to the hard structure of teeth in order to anticipate the claim?

No

This is an intended use recitation. The reference must be capable of applying an electrical field to the hard structure of teeth but does not have to disclose applying an electrical field to the hard structure of teeth.

Example 3:

Questions and Answers (*cont.*)

Q2: Must a reference disclose an electrode in order to anticipate the claim?

Yes

This is a positively claimed structural limitation.

Example 3:

Questions and Answers (*cont.*)

Q3: Must a reference disclose an electrode designed to be placed at a point on a patient's body in order to anticipate the claim?

No

However, the electrode must be disclosed and must be capable of being placed at a point on a patient's body.

Example 3:

Questions and Answers (*cont.*)

Q4: Must a reference disclose dental restorative materials or a layer of adhesive substance in order to anticipate the claim?

No

The reference must only be capable of being used with dental restorative materials and a layer of adhesive substance as claimed.

Example 3:

Questions and Answers (*cont.*)

Q5: Must a reference be capable of applying a restorative material in order to anticipate the claim?

Yes

Example 3:

Questions and Answers (*cont.*)

Q6: Must a reference be able to apply an adhesive substance at any particular efficiency or as well as the disclosed invention in order to anticipate the claim?

No

The claim merely requires an ability to apply but not at any particular level of efficiency.

Example 4: An Apparatus for Inspecting an Edge of Strip Metal

Video camera 41 views cut edge of strip and supplies signals to image processor 44 programmed to digitize the image and extract information defining the slit/shear ratio and straightness of the cut, information which is used by an operator to determine whether adjustment of the edge trimmers is required

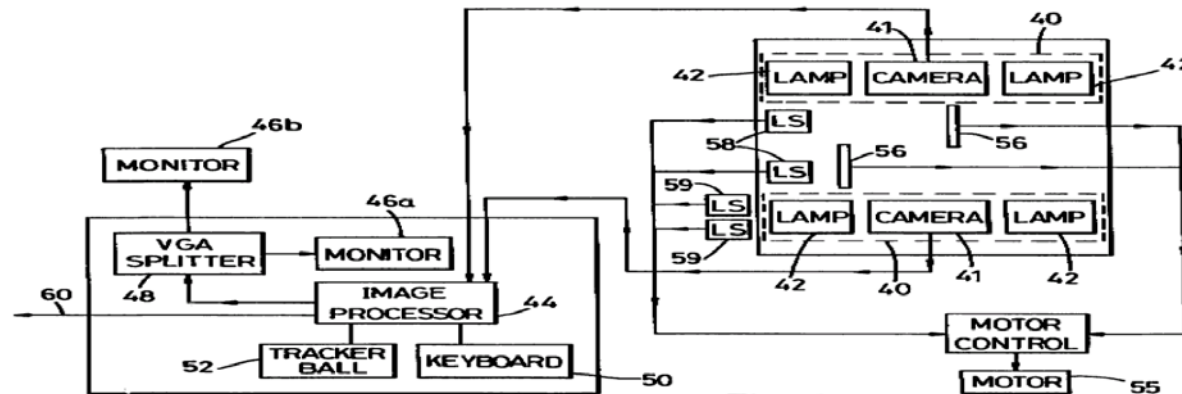


Fig. 4

Example 4:

Claim

An apparatus ***for inspecting an edge of strip metal after the edge has been trimmed by at least one cutting device***, the apparatus comprising:

- an inspection station through which the trimmed edge is passed, the inspection station comprising at least one image producer operable to form at least one image of the ***trimmed edge***;

- an image processor in operational association with the at least one image producer, the image processor operable to analyze the at least one image in real time ***to determine at least one quality characteristic of the trimmed edge***.

Example 4:

Questions and Answers

Q1: Does the claimed apparatus include a cutting device as part of the combination?

No

The cutting device is not positively recited.

Q2: Must an anticipating device disclose the process of inspecting an edge after it has been trimmed, as opposed to before it has been trimmed?

No

The anticipating device need only be capable of performing the intended use

Example 4:

Questions and Answers (*cont.*)

Q3: Is the claimed image processor limited to structure that determines at least one quality characteristic particular to trimmed metal?

No

It needs to be capable of functioning on trimmed metal and determining at least one quality characteristic of the trimmed edge.

Example 4: The Prior Art

United States Patent [19]
Fant et al.

[11] Patent Number: 4,519,041

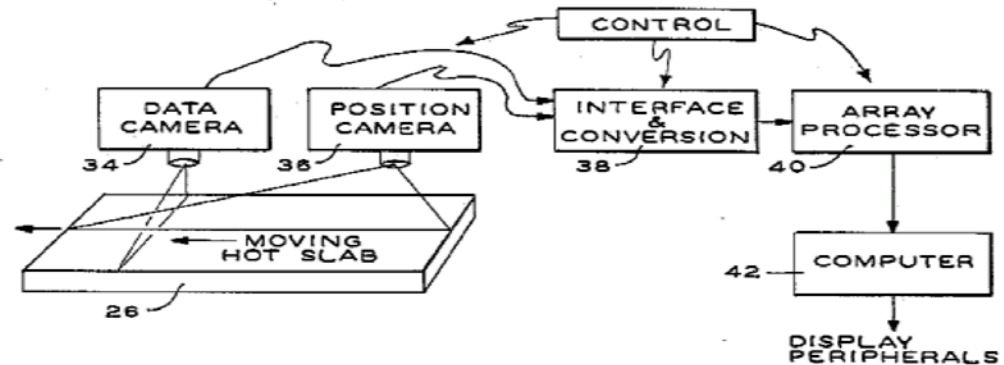
[45] Date of Patent: May 21, 1985

[54] REAL TIME AUTOMATED INSPECTION
[75] Inventors: Karl M. Fant, Minneapolis; Richard A. Fundakowski, Saint Paul; Ted S. Levitt, Minneapolis; John E. Overland, Plymouth; Bindiganavle R. Suresh, New Brighton; Franz W. Ulrich, Minneapolis, all of Minn.
[73] Assignee: Honeywell Inc., Minneapolis, Minn.
[21] Appl. No.: 374,373
[22] Filed: May 3, 1982
[51] Int. Cl.³ G06F 15/20; G06F 15/46; G06G 7/48
[52] U.S. Cl. 364/552; 364/469; 382/22; 382/54
[58] Field of Search 364/468, 469, 474, 476, 364/550, 552, 507, 472; 382/8, 16, 21, 22, 54; 356/429-431
[56] References Cited
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[57] **ABSTRACT**
A method and apparatus relating to the real time automatic detection and classification of characteristic type surface imperfections occurring on the surfaces of material of interest such as moving hot metal slabs produced by a continuous steel caster. A data camera transversely scans continuous lines of such a surface to sense light intensities of scanned pixels and generates corresponding voltage values. The voltage values are converted to corresponding digital values to form a digital image of the surface which is subsequently processed to form an edge-enhanced image having scan lines characterized by intervals corresponding to the edges of the image. The edge-enhanced image is thresholded to segment out the edges and objects formed by the edges are segmented out by interval matching and bin tracking. Features of the objects are derived and such features are utilized to classify the objects into characteristic type surface imperfections.

24 Claims, 43 Drawing Figures



Example 4:

Questions and Answers (*cont.*)

Q4: Should the prior art reference be used to reject the claim?

Yes

The prior art reference teaches a data camera capable of inspecting an edge of strip metal and is further capable of determining a quality characteristic of a trimmed edge.

Example 5: A Synthetic Sheet Material Claim

Claim 1. **A synthetic sheet material** comprised primarily of flash-spun plexifilamentary nonwoven fibers, said sheet material having a hydrostatic head pressure of at least about 75 cm of water, a Gurley Hill Porosity of less than about 15 seconds, and a moisture vapor transmission rate of at least 1300 g/m²/day.

Example 5:

Questions and Answers

Q1: Does the “synthetic sheet material” in the preamble limit the claim?

Yes

The prior art must be synthetic and must be a sheet.

Example 5:

Prior Art

The prior art reference discloses a synthetic sheet material which is composed of flash-spun plexifilamentary nonwoven fibers similar to the material disclosed in the instant specification but is silent regarding the claimed properties.

Example 5:

Questions and Answers (*cont.*)

Q2: Would an anticipatory prior art reference have to expressly teach the specific hydrostatic head pressure, Gurley Hill Porosity, and moisture vapor transmission rate, as claimed?

No

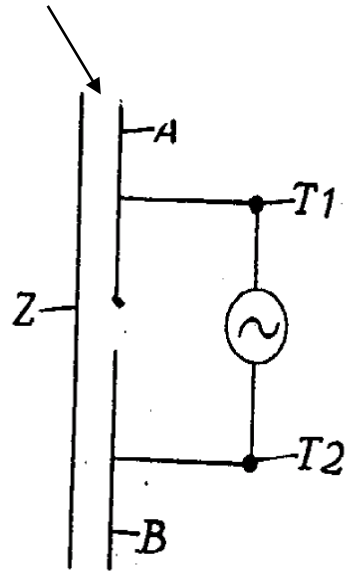
The prior art composition must inherently or expressly teach the claimed characteristics. If the examiner can show that the flash-spun plexifilamentary nonwoven fibers inherently possess the properties in the claim, the reference would be anticipatory.

Example 6: An Improved electroluminescent Lamp Claim

1. An Electroluminescent (EL) lamp comprising:
three electrodes, and
an EL material.
2. A clock comprising:
three electrodes, and
an EL material.

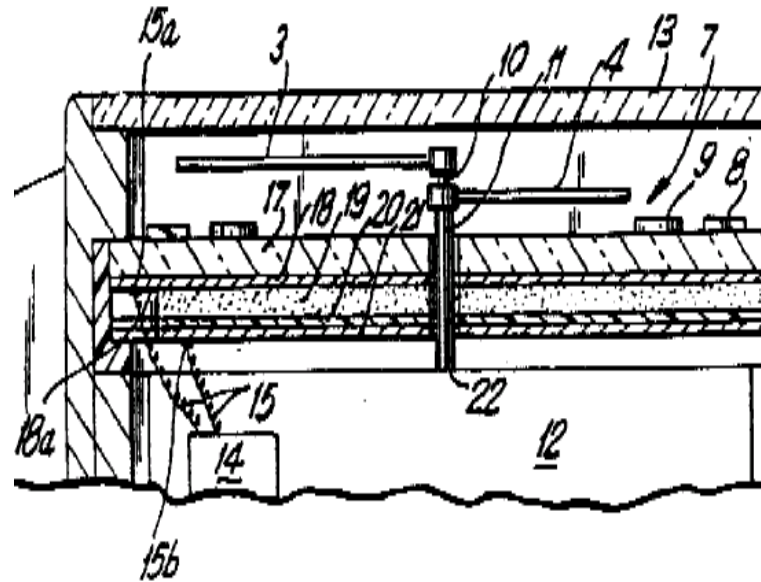
Example 6: Prior Art

EL Material (for Claim 1)



Electrodes A, B, Z

Clock (for Claim 2)



Example 6:

Questions and Answers

Q1: Does the recitation of “clock” in claim 2 impart structure or give “life meaning and vitality” to the claim?

Yes

You would not know that the body of the claim is in conjunction with a clock. Therefore, the preamble is necessary.

Example 6:

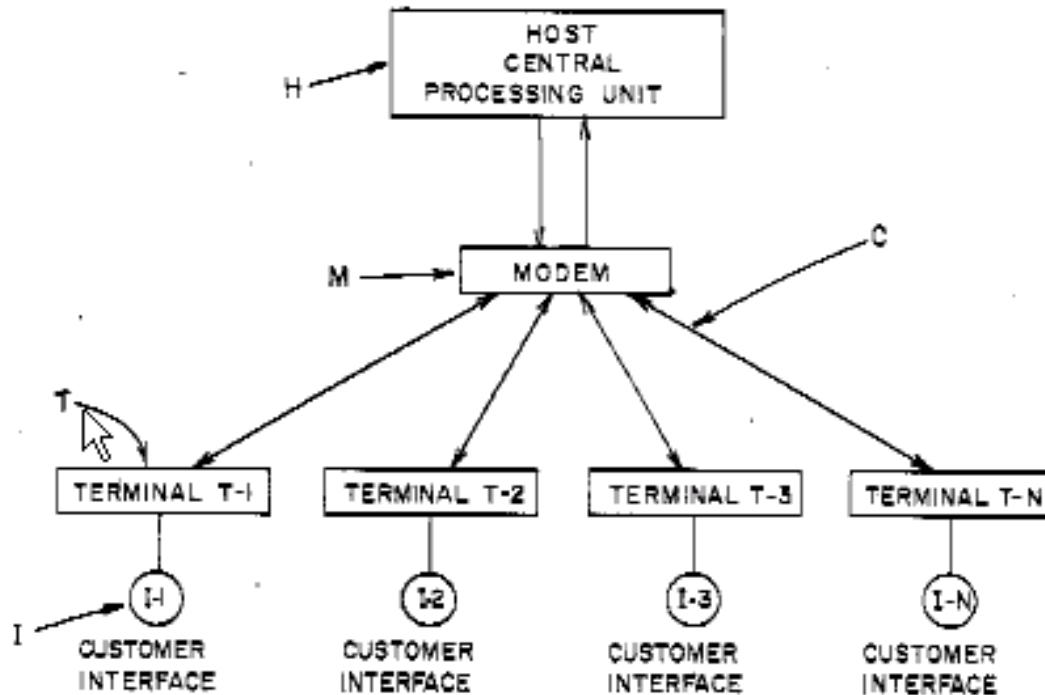
Questions and Answers (*cont.*)

Q2: Does the body of claim 1 and the body of claim 2 fully set forth all of the structural limitations of each claim?

No

In both claims, the body of each claim does not set forth all of the structural limitations of the claimed invention (lamp, clock). Therefore, each of the claim preambles must be treated as claim limitations.

Example 7: A System for Controlling the Selection and Dispensing of Product Coupons



Example 7:

Claim

A system for controlling the selection and dispensing of product coupons at a plurality of remote terminals **located at predesignated sites such as consumer stores**, wherein each of said terminals comprises:

activation means ...

display means

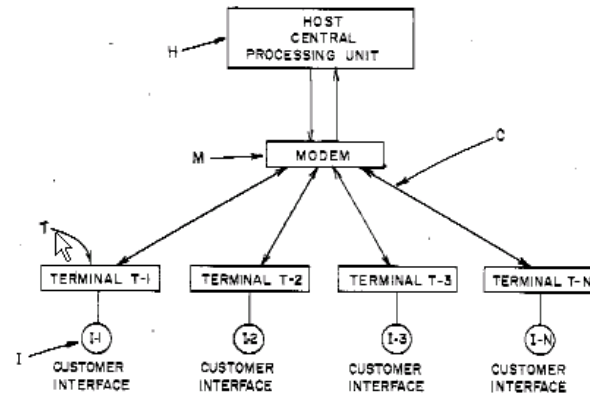
selection means ...

print means ...

control means ...

Example 7: Drawing

The drawing shows a system for controlling the selection and dispensing of product coupons, including a host central processing unit H connected to communicate with a modem M which in turn is connected to communicate with a plurality of remote terminals T-1 thru T-N, wherein each terminal includes a selection and printing device operated via a customer interface device I-1 thru I-N.



Example 7:

Question and Answer

Q1: Does the preamble phrase “located at predesignated sites such as consumer stores” limit the claim?

No

The apparatus must be capable of “controlling the selection and dispensing of product coupons at a plurality of remote terminals” and must be capable of being located “at predesignated sites.”

The “such as consumer stores” was not considered to be limiting as it is an example. The Court held that “‘Such as’ introduces an example of a broader genus rather than limiting the genus to the exemplary species.”

Example 7:

Question and Answer (*cont.*)

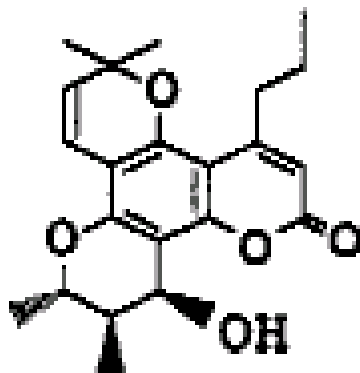
The Court also found that “located at predesignated sites such as consumer stores” in this situation does not limit the scope of the claim because:

- Applicant did not rely on this phrase to define the invention
- The phrase is not essential to understand limitations or terms in the body of the claim
- Deletion of the phrase from the preamble does not effect the structural definition or operation of the terminals
- The body of the claim defines a structurally complete invention
- The location of the terminals in stores merely gives an intended use for the claimed terminals

See: *MPEP 2111.02*

Example 8: A Cancer Therapeutic Composition Claim

1. A **cancer therapeutic composition** comprising a compound of structure A



and a pharmaceutically acceptable carrier.

Example 8:

Prior Art

- Reference A discloses a composition comprising a compound of structure A in a pharmaceutically acceptable carrier.
- Reference A teaches that the composition is used as an antiviral therapeutic for treating human immunodeficiency virus type 1 (HIV-1) infections.

Example 8:

Question and Answer

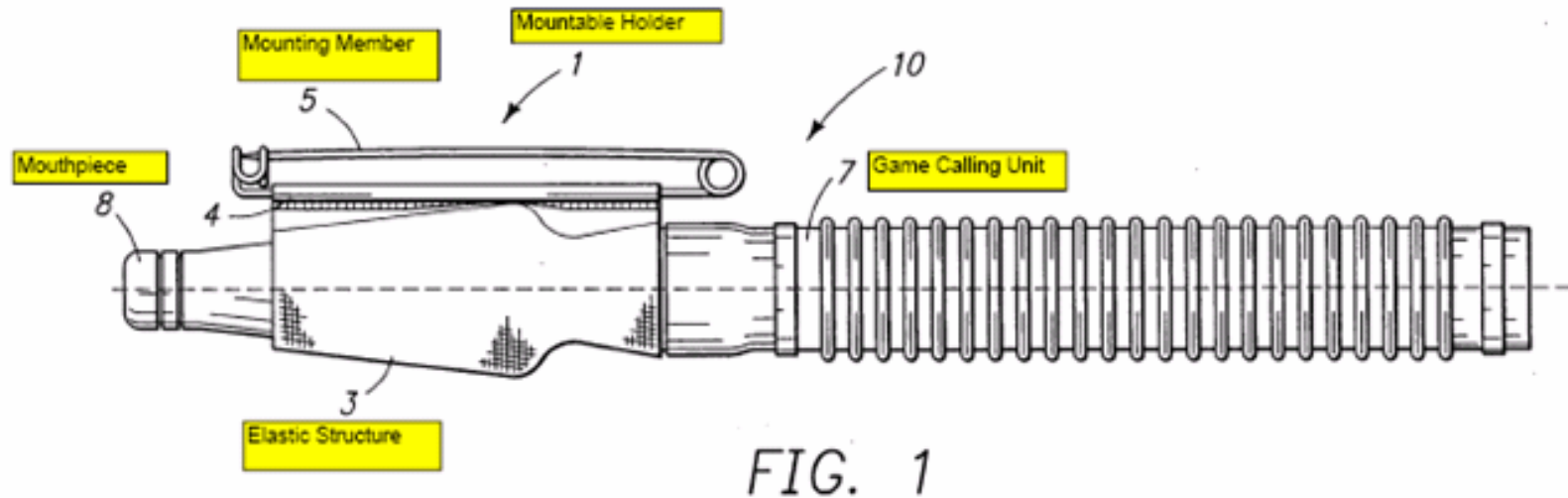
1. Does the prior art support a rejection?

Yes

- The compound and composition found in the prior art and in the instant composition are identical.
- The preamble of the claim (cancer therapeutic composition) merely recites an intended use of the composition and as such does not limit the claims.
- Therefore, the prior art composition anticipates the claimed composition.

See: *MPEP 2111.02*

Example 9: A Mountable Game Calling Assembly



Example 9:

Claim

A mountable game calling assembly **equipped to mount onto an upper arm garment of a game hunter and retain a detachable game calling unit at a predetermined aiming mount position so as to allow the game hunter to simultaneously call a hunted prey while positioning an aiming arm and an aimed weapon at the hunted prey**, said mountable assembly consisting essentially of the game calling unit equipped with a mouthpiece for calling the hunted prey, a mountable holder having an elongated elastic sleeve for elastically holding and retaining the calling unit in longitudinal alignment with the aiming arm and a mounting member equipped for longitudinally mounting and aligning the game calling assembly onto the upper arm garment of the aiming arm so as to thereby allow the game hunter to simultaneously call and aim the aimed weapon at the hunted prey when said calling assembly is thereby mounted onto said upper arm garment of said aiming arm at the predetermined aiming mount position.

Example 9:

Question and Answer

Q1. What's a prior art reference required to disclose with regard to the preamble language?

The prior art reference must be capable of:

- being mounted onto an upper arm garment of a game hunter
- retaining a detaching game calling unit at a predetermined aiming mount position
- allowing a game hunter to simultaneously call a hunted prey while aiming a weapon at the hunted prey

Example 10: Vaccine Claim

1. A **vaccine** comprising an isolated protein comprising SEQ ID NO:1 or a portion thereof which is antigenic.

Example 10:

Meaning of Vaccine

- *Merck Manual of Diagnosis and Therapy* (16th ed. 1992), p. 21
 - A suspension of whole or fractionated microorganisms that have been rendered non-pathogenic, given to induce an immune response and prevent subsequent disease
- *Dorland's Medical Dictionary* (25th ed. 1974)
 - a suspension of attenuated or killed microorganisms administered for the prevention, amelioration, or treatment of infectious diseases

Example 10:

Question and Answer

Q1: A prior art reference teaches the claimed composition comprising the recited protein in a pharmaceutically acceptable carrier. However, the prior art compound decreases immunity and makes one susceptible to subsequent disease. Does the prior art compound anticipate the claimed vaccine?

No

The recitation of “vaccine” in the preamble breathes life and meaning into the claim. A vaccine must increase immunity and prevent subsequent disease (e.g. a protective immune response).

Summary

You should now better understand how a preamble can limit the scope of a claim

Questions?



Please send your questions to
STEPPQUESTIONS@USPTO.GOV

Thank You!

Please Complete the Survey!

